YantraShare

Fair, accountable sharing of idle crop-residue machinery at Custom Hiring Centres

Team: Code Crafters
Problem statement: Unlock Underused Equipment and Spaces
Live product: **[FILL: public URL on Render]**
Source repository: **[FILL: GitHub URL]**

1. Team details and member roles

Every member also submits individually with their own role and contribution.

Member	Role	What this person personally built
[FILL: Name 1]	Backend / Team lead	Database models, authentication and roles, booking state machine, audit log, Render deployment.
[FILL: Name 2]	Allocation and recovery engineering	Scoring and ranking with plain-language explanations, usage caps, waitlist, breakdown and overdue recovery, notifications.
[FILL: Name 3]	Frontend / UI-UX	HTML and CSS templates, farmer request and status pages, owner dispatch board, notification bell.
[FILL: Name 4]	Product, research, QA and documentation	Field interviews, evidence collection, seed data, condition-check forms, testing, this document.

2. Selected use case and understanding of the problem

Use case

- Shared crop-residue management (CRM) machinery, such as Super Seeders and Happy Seeders
- Held by Custom Hiring Centres (CHCs): cooperative societies, panchayats, farmer producer companies, registered farmer societies and individual farmer-entrepreneurs
- Focus on Punjab, where the evidence base is strongest
- The workflow applies to any institution that lends equipment under conditions

Why this is a sharing problem, not just a booking problem

- India has built the supply side: government schemes have subsidised machines and created tens of thousands of CHCs
- The weak point is day-to-day operation: who gets a machine when several farmers want it in the same short harvest window, what happens when it breaks or comes back late, and what condition it was in before and after use
- These are the trust, safety and accountability requirements in the problem statement
- Today they are handled by phone calls, personal judgement and handwritten registers

Context

- Paddy is harvested between September and November in Punjab
- Machines are needed intensively for a few weeks and sit idle for most of the year
- A request that cannot be served in time can push a farmer toward burning residue, which harms air quality and soil health

3. Target users

User	Need	How the product serves them
Farmer (small, marginal, medium)	Reliable access to a machine at the right time and an affordable rate; understand why a request was accepted or delayed	Simple request form, eligibility checklist, status page with a plain-language reason, notification if the machine changes
CHC owner / operator	Know who has which machine, avoid damage disputes, keep machines earning, decide fairly under pressure	Dispatch board with conflicts highlighted, ranked suggestions the owner can approve or override, condition records, history
Manager (society staff)	Run day-to-day approvals within the owner's rules	Role-based access to review, approve and record returns
Administrator / official	See whether access is fair and machines are used	Usage and fairness dashboard, full audit trail, incident log

4. Evidence that the problem matters

- **Primary source:** CEEW report *Custom Hiring Centres for Better Crop Residue Management: Insights from Punjab* (Jain, Kemanth and Sarao, September 2025)
- **Basis:** in-depth interviews with 65 CHC operators in Ludhiana, Patiala and Sangrur, September to November 2024
- **Figures below** are from that report unless another source is named

Finding	Evidence	What it means for our design
Machines exist but are underused	CHCs own about 45% of all CRM machines in Punjab, yet only about 15% of farmers practising in-situ CRM rent through them. About 53% of renting CHCs have at least one underutilised machine, almost always the Happy Seeder.	Access and scheduling matter, but obsolescence also matters. We do not claim software alone fixes idle machines.
Digital tools have low uptake	Only 16% of renting CHCs had heard of i-Khet and none had used it. About 13% knew Unnat Kisan and about 3% had used it. Roughly 1% of Punjab farmers use such apps (earlier CEEW survey of about 1,500 farmers).	Do not require every farmer to install an app. Let the operator log phone requests and give farmers a simple status link.
Work runs by phone and in person	Almost all CHCs take requests in person or by phone. About 10% use WhatsApp groups. All keep manual records, and 95% collect only verbal feedback that is never written down.	Design around the phone call and create a written, searchable record automatically.
First-come, first-served, with in-group priority	Most CHCs allocate first-come, first-served. Roughly half of private CHCs use machines only for their own group despite an 80% subsidy.	Transparent rules and usage limits, so access is not decided by who knows the operator.
Operators already use urgency in practice	A cooperative society secretary described giving the machine to the farmer whose field cannot wait, when two arrive together.	We formalise this real rule as urgency-based ranking, with the reason shown to everyone.
Damage and late returns are common	About 77% of renting CHCs report damage from mishandling and 45% report late returns. Damage is often found only after return, and 64% of affected CHCs pay for repairs themselves.	Before-and-after condition checks, overdue alerts and incident records.

Finding	Evidence	What it means for our design
Machines are scarce at peak	An ICAR-NDRI study of 180 farmers using CHCs in Punjab found farmers wanted more of the high-demand machines in societies during peak season.	Waitlists and recovery when a machine is unavailable.
Money is at stake	A packaged service (machine, fuelled tractor, driver) costs about INR 2,000 per acre. A Super Seeder needs roughly 105 acres a year to break even, and only about 36% of surveyed CHCs reached that.	An owner break-even tracker gives owners a business reason to lend machines.

Scale

- The CRM Scheme has supported roughly 40,000 CHCs across Punjab, Haryana and Uttar Pradesh
- Punjab alone has about 25,400 CHCs and 1.37 lakh machines
- Around INR 1,756 crore was received under the scheme in Punjab

Limits of this evidence

- The CEEW study interviewed CHC operators, not farmers
- Responses were self-reported, with possible social-desirability and recall bias
- Sampling was purposive and convenience-based in three districts

5. Existing solutions and the specific gap

We do not claim that nobody has built agricultural equipment sharing.

- **FARMS (Farm Machinery Solutions):** an official multi-language mobile app that lists CHCs, farm machinery banks and hi-tech hubs for hiring, lets individual farmers list machines for rent, and supports sale and purchase of old machinery
- **i-Khet and Unnat Kisan:** earlier government apps for listing and booking CRM machines, with the low adoption described above
- **CS Posible:** a pilot site where cooperative societies upload rental data; other CHC types have no equivalent
- **Other models:** Hello Tractor (Africa), Krish-E and Raj Kisan CHC (India), mentioned in the CEEW recommendations

The gap we address

- Listing and booking show that a machine exists
- They do not resolve what happens when several farmers need the same machine at the same time
- They do not explain the decision, discourage monopolisation, or record condition and returns
- They do not recover when a machine becomes unavailable
- On the public FARMS description we found no mention of conflict resolution, explainable allocation or breakdown recovery

6. Proposed solution

A lightweight web platform that sits on top of how CHCs already work: an operations and accountability layer, not a marketplace farmers must join.

- **Owner-first dispatch.** The operator logs requests, including phone calls, in seconds. Farmers track status through a simple link.
- **Explainable allocation.** Competing requests are ranked by urgency (days the field can wait), small and marginal farmer priority, recent usage, then request time. Every farmer sees the reasons. The owner can override with a recorded reason.
- **Recovery from unavailability.** If a machine is marked broken or overdue, affected bookings are re-ranked, the next suitable machine or nearby centre is suggested, and farmers are notified. A human confirms the change.
- **Condition and trust records.** Before-and-after checklists with optional photos, damage and delay reports, and a full audit history for both parties.
- **Fair-usage limits.** Caps on repeated bookings by one person or group so limited machines are shared.

7. Complete product workflow

Step	What happens	Role
1. List	Owner adds a machine with type, tractor horsepower required, rate, location, availability windows, usage rules and eligibility conditions.	Owner
2. Discover	Farmer searches by machine type, acreage and date and sees an eligibility checklist.	Farmer
3. Request	Farmer submits acreage, field location, days the field can wait and whether a tractor and driver are needed. Owners can also log a phone request.	Farmer / Owner
4. Verify	System checks listed requirements (farmer category, tractor horsepower, operator, usage cap). Document verification is a labelled mock approved by the owner.	System
5. Review	Owner or manager sees ranked requests with explanations and any conflicts.	Owner / Manager
6. Decide	Request is approved, rejected or waitlisted. The farmer sees the reason.	Owner / Manager
7. Check-in	Initial condition checklist and optional photo recorded at handover.	Owner + Farmer
8. Return	Final condition checklist recorded and compared with the initial one.	Owner + Farmer
9. Exceptions	Late return, damage, breakdown or conflict is reported and handled, with recovery suggestions and notifications.	All
10. History	Both parties view a time-ordered audit trail of every decision and action.	All

8. Features and build status

Feature	Status	Notes
Resource listings with availability, location and usage rules	Planned	Owner sets machine type, tractor HP needed, rate, location, availability and rules
Search and filtering by machine type, date and acreage	Planned	Farmers find a suitable machine quickly
Role-based access (farmer, owner, manager, admin)	Planned	Each role sees only its own pages and actions
Eligibility checks (category, tractor HP, operator, usage cap)	Planned	Rules checked automatically when a request is made
Booking requests with approve, reject and waitlist	Planned	Owner decides and the farmer sees the reason
Transparent ranking of competing requests with plain-language reasons	Planned	Key differentiator: urgency, farmer category, recent usage, request time
Owner override with mandatory recorded reason	Planned	Human oversight: every override is logged
Check-in and check-out with before-and-after condition reports	Planned	Same checklist at handover and return, compared side by side
Damage, delay and incident reporting	Planned	Reports linked to the booking and the machine history
Breakdown and overdue recovery with re-ranking	Planned	Key differentiator: queue re-ranked and a replacement suggested
In-app notifications (approval, reminder, cancellation, overdue)	Planned	Notification bell with unread count
Fair-usage caps against monopolisation	Planned	Limits repeat bookings by one person or group
Auditable activity history for every important action	Planned	Approvals, overrides, check-ins, returns and reports are all logged
Phone-request intake with farmer status link	Planned	Stretch: operator logs a call and the farmer gets a status link
Machine condition history and owner break-even tracker	Planned	Stretch: condition history, acres covered and break-even per machine
Fairness dashboard for administrators	Planned	Stretch: usage share by person and group, flagged for human review

9. Technology stack and architecture

Layer	Choice	Reason
Frontend	HTML5 and CSS3 (custom stylesheet) rendered from Jinja2 templates, with small amounts of HTMX / JavaScript for inline actions	No separate frontend build, loads quickly on phones, easy for the whole team to edit
Backend	Python with FastAPI	Small, fast to build, clear routes and validation
Database	PostgreSQL through SQLAlchemy	Persistent storage on the hosting platform; relational model suits bookings and audit trails
Authentication	Session cookies, bcrypt password hashing, role checks on every route	Role-based access without external services
Audit and notifications	Two database tables written by every state change	Complete, queryable history
Hosting	Render web service connected to the Git repository	Simple deployment from source
Version control	Git and GitHub	Commit history shows the product was built in the hackathon window

Architecture

- Browser to FastAPI application (routes, role checks, booking state machine, allocation module) to PostgreSQL
- A booking moves through: requested, approved / rejected / waitlisted, checked-in, returned, closed
- Each transition writes an audit event with actor, action, reason and time

AI and external services

- Allocation uses rule-based scoring with plain-language reasons, not a black-box model
- No access, safety or penalty decision is made by an unexplained automated system

Disclosure of existing components

- Libraries and frameworks: FastAPI, SQLAlchemy, Jinja2, bcrypt, HTMX
- Open-source libraries only; no pre-built templates or existing project code
- AI assistant: Claude, used for research support and drafting this document

10. Trust, safety, privacy and fairness

- **Human control.** The system recommends and explains. The owner or manager approves, rejects or overrides, and every override needs a recorded reason.
- **Explainability.** Each farmer sees the rules and factors behind their approval, rejection or waitlist position.
- **Verification.** Eligibility rules are checked automatically where possible. Document checks in the prototype are a labelled mock that an owner approves manually.
- **Permissions.** Farmers see only their own bookings, owners see their own machines, admins see aggregate and audit views.
- **Privacy.** Only data needed for a booking is collected. Status links use random, expiring tokens. Demonstration data is fictional.

- **Fairness.** Usage caps and waitlists discourage monopolisation. Small and marginal farmers receive priority, consistent with Punjab guidance that they should pay operational charges only. Flags from the fairness dashboard go to a person and never trigger automatic penalties.
- **Accountability.** Condition records and a complete audit trail support dispute handling for owners and users.

11. Testing notes for judges

Role	Username	Password
Farmer A	farmer_a	Demo@1234
Farmer B	farmer_b	Demo@1234
Owner / Operator	owner	Demo@1234
Manager	manager	Demo@1234
Administrator	admin	Demo@1234

Suggested journeys

- **Journey 1, full lifecycle:** log in as Owner and list a machine. As Farmer A, find it, request it and see the eligibility result. As Owner, approve. Record check-in, then return with a condition report. Open the history to see every step.
- **Journey 2, competing requests:** Farmer A and Farmer B request the same machine and dates. As Owner, open the dispatch board and view the ranking and reasons. Approve one and see the other waitlisted with an explanation.
- **Journey 3, recovery:** mark the approved machine as broken. See the queue re-ranked, a replacement suggested and notifications sent. Confirm as Owner.
- **Journey 4, accountability:** report damage on a return, then open the audit history as Admin.

Notes

- The free hosting tier can take about a minute to wake after inactivity, so please allow time on first load

12. Known limitations

- Document and training verification is simulated and needs integration with real institutional records
- Notifications are in-app only; SMS or WhatsApp delivery is not implemented
- Allocation weights are a hypothesis based on published evidence and need field testing with operators
- The platform does not solve machine obsolescence, spare-parts shortages or driver shortages, which the CEEW report identifies as separate problems
- Evidence comes mainly from Punjab CHC operators and may differ in other states

13. Resources and references

Primary evidence

- Jain, S., Kemanth, K. and Sarao, N. S. (2025). *Custom Hiring Centres for Better Crop Residue Management: Insights from Punjab.* New Delhi: CEEW. Open access, CC BY-NC 4.0. https://www.ceew.in/publications/improving-access-to-crop-residue-management-solutions-with-custom-hiring-centres-in-agriculture (full PDF: https://www.ceew.in/sites/default/files/ceew-custom-hiring-centres-for-web.pdf)

- Kemanth, K. et al. (2024). CEEW farmer survey of about 1,500 farmers, *How can Punjab Increase the Adoption of Crop Residue Management Methods?* Cited in the 2025 report for the 15% access and 1% app-use figures.
- ICAR-NDRI, Karnal (2020). *Mitigation of Stubble Burning through Custom Hiring Centres in Punjab* (180 CHC-using farmers). https://krishikosh.egranth.ac.in/items/0a5aa5d4-9dbc-457a-87b2-2450170431e4

Existing platforms

- Department of Agriculture and Farmers Welfare, Government of India. FARMS: Farm Machinery Solutions. https://agrimachinery.nic.in/Index/farmsapp
- i-Khet and Unnat Kisan apps, and CS Posible (India Paryavaran Sahayak Foundation), as described in the CEEW report

Policy and rules referenced

- Crop Residue Management Scheme and its operational guidelines (2018 onward, including 2024 amendments), Ministry of Agriculture and Farmers Welfare, as summarised in the CEEW report
- Punjab Department of Agriculture and Farmers Welfare (2024) fixed hiring charges for CHC machines, as tabulated in the CEEW report

Technology documentation

- FastAPI, SQLAlchemy, Jinja2, HTMX and Render official documentation

All demonstration data in the product is fictional. No confidential or personal records were used.